whole structure looks like inverted letter 'Y' and is called **translator**. When the insect visits the flower for nectar, the translator gets attached to the proboscis or leg of the visitor. During the visit to the next flower the pollinia come in contact with the receptive stigma carrying out pollination.

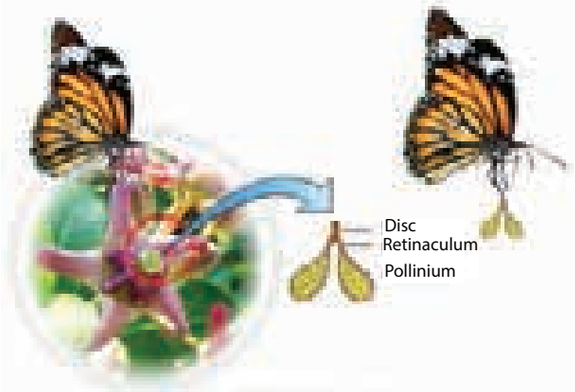


Figure 1.17 (b) Pollination in *Calotropis* -Translator mechanism

Pollination in *Aristolochia* (Trap mechanism)

A special mechanism to accomplish pollination called trap mechanism is found in *Aristolochia*. The flowers are axillary and perianth is tubular with a hood at the top. The basal region is swollen and possesses gynostegium. The gynostegium has six anthers.

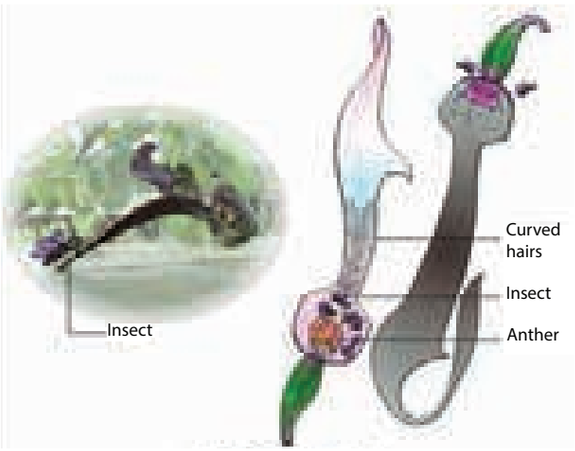


Figure 1.17 (c) Pollination in *Aristolochia* - Trap mechanism

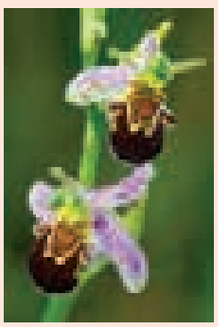
The inner wall of tubular middle part of the perianth is slippery and lined with stiff hairs which are pointed downwards. The young flowers are erect. During this stage small flies



Pollination – A composite event

Pollination provides information about evolution, ecology, animal learning and foraging behaviour. Flowers not only supply nectar but also provide microclimate, site and shelter for egg laying insects. The association of insects benefits the flower by getting pollinated and ensures the propagation of its own progeny. The floral parts are well modified in shape, size to attract the pollinators to accomplish pollination .

The relationship between *Yucca* and moth (*Tegeticula yuccasella*) is an example for obligate mutualism. The moth bores a hole in the ovary of the flower and lays eggs in it. Then it collects pollen and pushes it in the form of balls down the hollow end of the stigma. Fertilization takes place and seeds develop. Larvae feed on developing seeds. Some seeds remain unconsumed for the propagation of the plant species. It is interesting that the moth cannot survive without *Yucca* flowers and the plant fails to reproduce sexually without the moth.



Bee Orchid

Similarly in *Amorphophallus*, flowers apart from providing floral rewards, also forms safe site for laying eggs. Many visitors consume pollen and nectar and do not help in pollination. They are called pollen / nectar robbers.

In Bee orchid (*Ophrys*) the morphology of the flower mimics that of female wasp (*Colpa*). The male wasp mistakes the flowers for a female wasp and tries to copulate. This act of pseudocopulation helps in polli nation.

The pollination in Fig (*Ficus carica*) by the Wasp (*Blastophaga psenes*) is also an example for similar Plant – insect interaction.

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